

E-commerce Cohort Retention & CLTV Analysis

Full-stack Data Analytics Project

Submitted By

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Submitted To

INFOTACT SOLUTIONS

Project Description

This project presents a Full Stack End-to-End Solution for E commerce Cohort Retention and CLTV Analysis using Python and Power Bi for analyzing cohort retention, CLTV, Customer Segmentation, Order Frequency and Average Order Value. This Project was carried out with Olist E-commerce Brazilian Dataset which was used to build one master transaction table for analysis and deriving key business metrics. The project combines data preparation, exploratory analysis, customer segmentation, cohort retention analysis, CLTV analysis, and interactive Power BI dashboards.

Technology Stack

- Python (pandas, numpy, matplotlib)
- Microsoft Power BI & DAX
- GitHub

Key Business Metrics

- Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value
 - Purchase Frequency
 - Repeat Rate
 - Customer Lifespan
 - Historical CLTV
 - Segment Level Customer Contribution
 - Retention Rate and Matrix
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Project Domain

Customer Retention & CLTV Analytics

Project Timeline

26th July 2026 to 24th August 2026

1. Project Overview

The Project is built as part of Data Analytics Internship project with Infotact Solutions. The project focuses on analyzing Olist Brazilian E-commerce dataset to derive business insights like Purchasing behaviour, historical CLTV, cohort retention analysis, Repeat rate etc.

This project analyzes customer purchasing behavior, retention patterns, customer segmentation, and Customer Lifetime Value (CLTV) using Python and Microsoft Power BI.

The objective is to transform transactional customer data into actionable business insights that help answer questions such as:

- How effectively are customers being retained over time?
- Which customer segments generate the most value?
- How does customer purchasing behavior differ across segments?
- Which customers contribute the highest CLTV?
- How do purchase frequency, order value, and customer lifespan influence customer value?

The project combines data preparation, exploratory analysis, customer segmentation, cohort retention analysis, CLTV analysis, and interactive Power BI dashboards.

The tools and technologies used to built this project are : Python for EDA and building master transactional table for analysis. Python libraries like pandas, Numpy and Matplotlib were used for exploratory data analysis, cohort retention and CLTV analysis and visualizing various charts like revenue over month, customer segmentation, delivery delays etc respectively.

For Data Modelling and Dashboard, Microsoft Power BI with DAX and Power Query Features were used. Data from raw dataset and master transaction table was loaded to power BI and Start Schema was built for efficient analysis filtering. Multiple Dimension Tables were build with Fact table for creating a star schema.

Finally as an output for the project two dashboards were build for executive overview of purchasing behaviour, cohort matrix and for historical CLTV analysis.

2. Business Problem

A business may generate substantial revenue while still facing challenges such as:

- Low repeat-purchase behavior
- Declining customer retention over time
- Unequal contribution of different customer segments
- Limited understanding of customer lifetime value
- Difficulty identifying high-value customers
- Low purchase frequency among a large proportion of customers

This project addresses these challenges by analyzing the customer journey from first purchase → repeat purchase → retention → lifetime value.

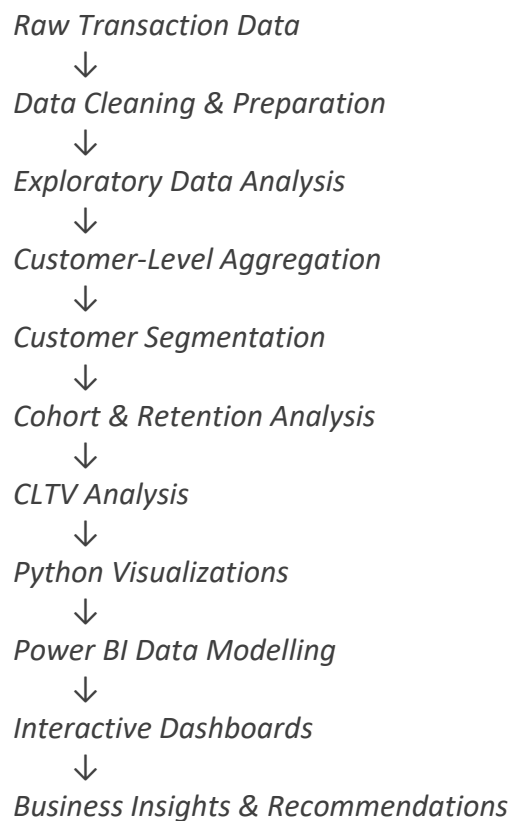
Key Business Questions Answerd

1. How is customer retention changing across acquisition cohorts?
2. Which customer segments contribute the most revenue and CLTV?
3. What differentiates high-value customers from medium- and low-value customers?
4. How do Average Order Value and purchase frequency relate to customer value?
5. Which customers have the highest CLTV and therefore represent the greatest value to the business?
6. Where should the business focus its retention and customer-value improvement efforts?

3. Tools & Technologies

Tools	Purpose
Python	Data cleaning, transformation, exploratory analysis and analytical modelling
Pandas	Data manipulation and preparation
Numpy	Numerical calculations
Matplotlib / Seaborn	Exploratory data visualization
Jupyter Notebook	Python-based analysis
Power BI	Interactive dashboards and business reporting
DAX	Measures, KPIs and analytical calculations
GitHub	Project documentation and version control

Project Workflow



4. Data Preparation & Cleaning

The transactional data was prepared for customer-level and time-based analysis.

The Olist Brazilian E-commerce dataset included multiple tables like customers table, orders table, payments table, products table, order items table and product names translation table. All the tables were loaded in Jupyter Notebook and were cleaned and standardized for better analysis and pulling out business insights. All tables were then merged in a single master transaction table for star schema preparation.

Main preparation activities

- Checked and handled missing values
- Reviewed duplicate records
- Validated data types
- Converted transaction dates into usable date fields
- Created customer-level aggregations
- Derived order and purchase metrics
- Prepared cohort-related date attributes
- Prepared variables required for customer segmentation
- Created analytical measures required for CLTV and retention analysis
- Removed unnecessary columns

The cleaned data was then used for both Python analysis and Power BI reporting.

5. Python Analysis

Python was used to perform the analytical groundwork before building the Power BI dashboards.

Major areas analyzed were :

A. Cohort Retention Analysis

Customers were grouped according to their acquisition date/first-purchase month to form a cohort group which was then used for building cohort retention matrix and retention analysis.

Retention was then evaluated across subsequent periods to identify:

- Retention decline over time
- Differences between customer cohorts
- Strength of repeat purchasing
- Long-term customer engagement patterns

Cohort Retention Matrix was calculated using following steps:

- Creating a data frame for cohort analysis
- Finding each customers first purchase date/acquisition date
- Calculating cohort index (month since first purchase)
- Building the cohort table consisting of cohort month, index and customer key
- Building cohort retention matrix using *pivot()* function
- Visualizing the matrix in the form of heat-map to derive insights.



B. Customer Lifetime Value Analysis

One of the main objective of the product was to calculate Historical CLTV and Customer Analysis. CLTV analysis was performed to understand the economic value generated by customers over their relationship with the business.

The analysis considered customer purchasing behavior such as:

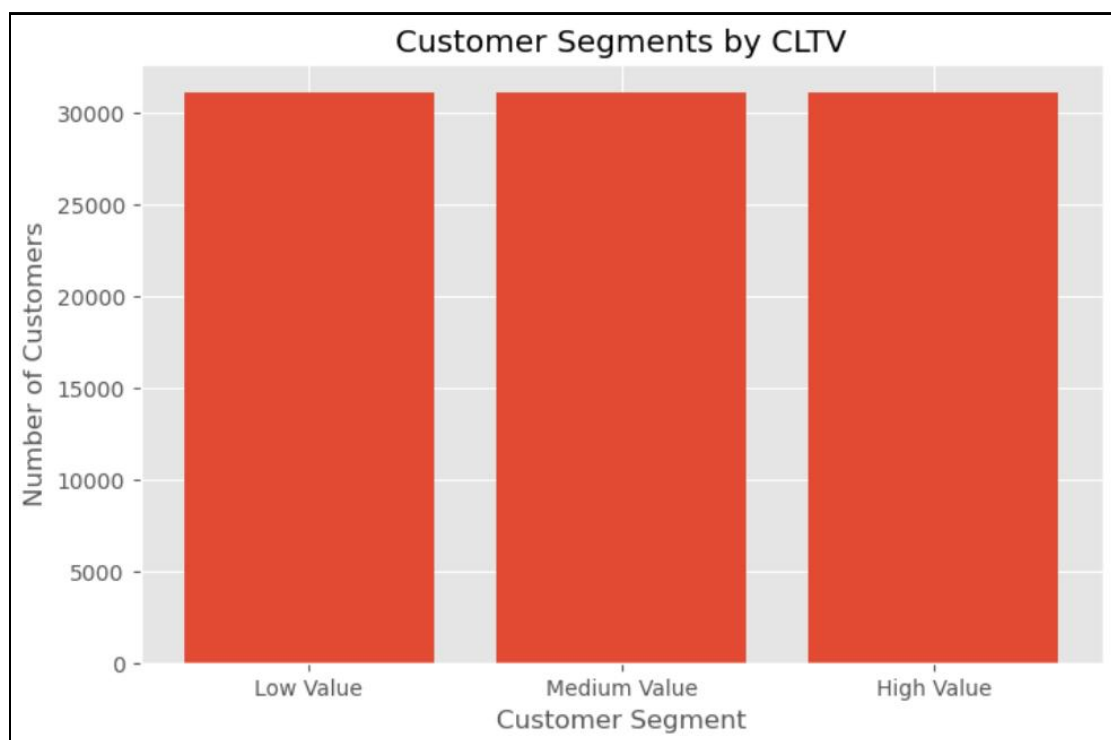
- Revenue
- Purchase frequency
- Average Order Value
- Customer lifespan
- Repeat purchasing

C. Customer Segmentation

Customers were segmented based on their historical CLTV and their value to the business.

Customers were divided into three value-based segments:

- High Value
- Medium Value
- Low Value



6. Data Modeling and Star schema

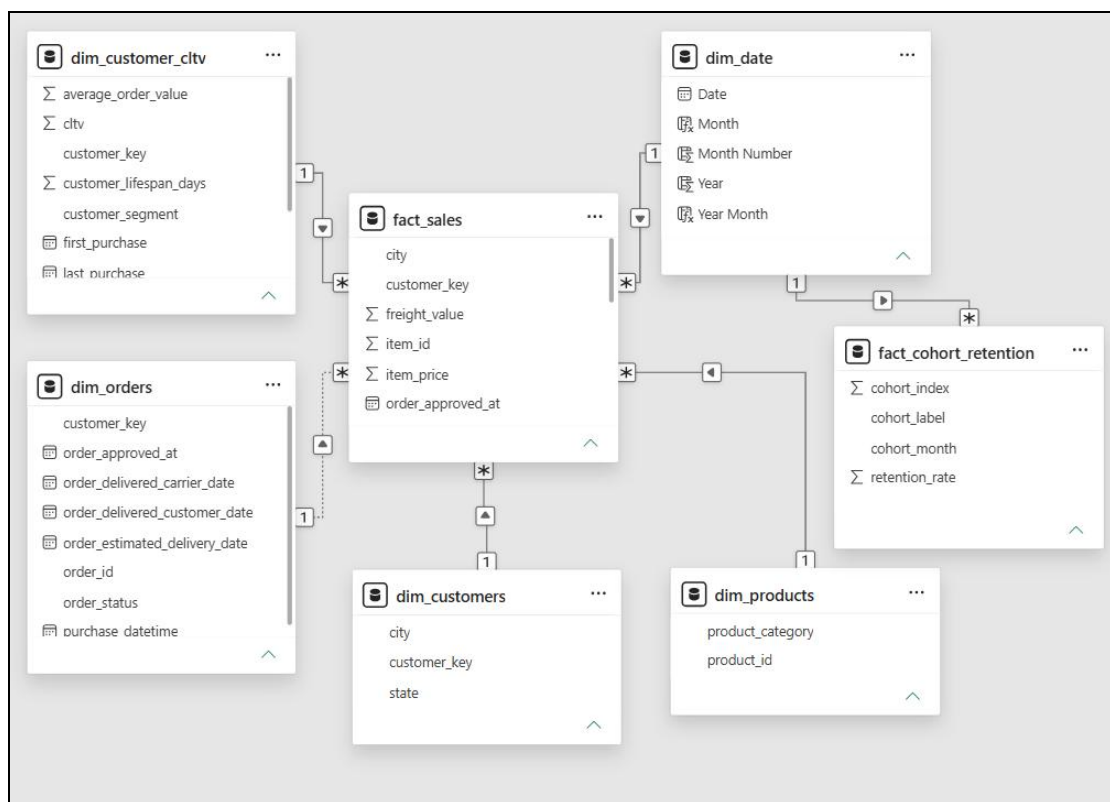
The Master Transaction Table `fact_sales` and other cleaned and per-processed tables were then loaded to Power BI and Data Model was built using Star Schema Architecture.

The analytical data was modeled in Power BI to support interactive filtering and dashboard reporting.

The following Fact and Dimension tables were build such as:

- `fact_sales`
- `dim_customers`
- `dim_products`
- `dim_orders`
- `dim_customer_cltv`
- `dim_date`
- `fact_cohort_retention`

Relationships were build between the tables for integration and filtering. Interactive filters allow users to explore the analysis by relevant business dimensions.



7. Power BI Dashboards

Dashboard 1 — Cohort Retention Overview

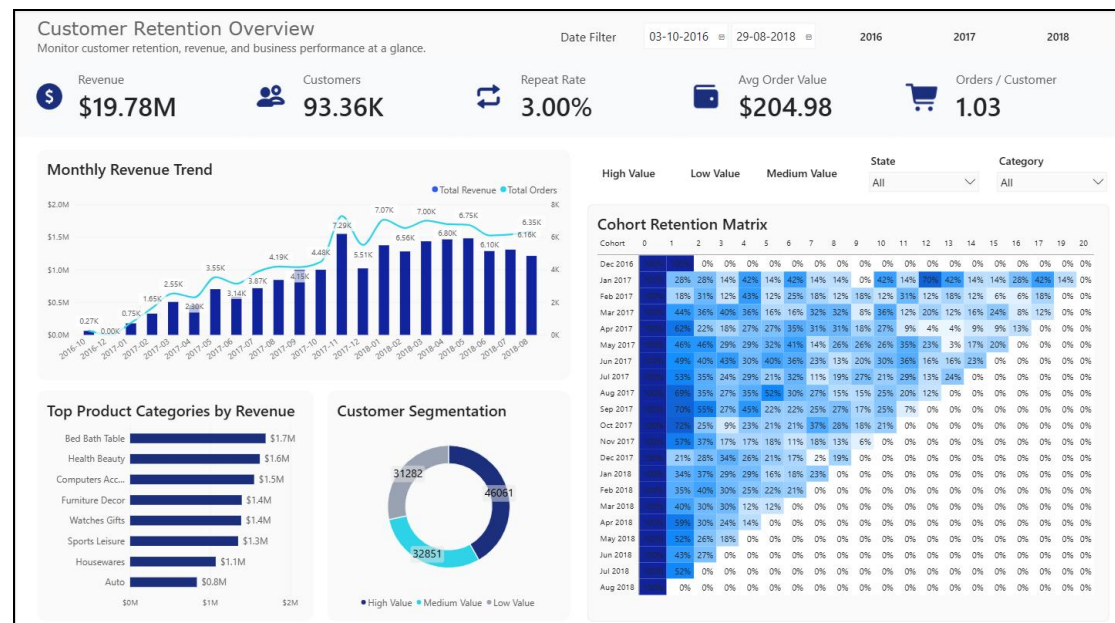
This dashboard focuses on overall customer performance and retention. It monitors overall customer retention, revenue and business performance at glance.

Main KPIs

- Revenue: \$19.78M
- Customers: 93.36K
- Repeat Rate: 3.00%
- Average Order Value: \$204.98
- Orders / Customer: 1.03

Main visuals

- Monthly Revenue & Orders Trend
- Cohort Retention Matrix
- Top Product Categories by Revenue
- Customer Segmentation
- Interactive State and Category filters



Key observations

- The business generated approximately \$19.78M in revenue from around 93.36K customers.
- Overall repeat purchasing is relatively low, with a 3% repeat rate.
- Average order value is approximately \$204.98.
- Purchase behavior is close to one order per customer, reflected by 1.03 orders per customer.
- The cohort matrix shows that retention generally decreases as customers move further away from their initial purchase period.
- Customer value is unevenly distributed across the three customer segments.

Dashboard 2 — Customer Lifetime Value Analysis

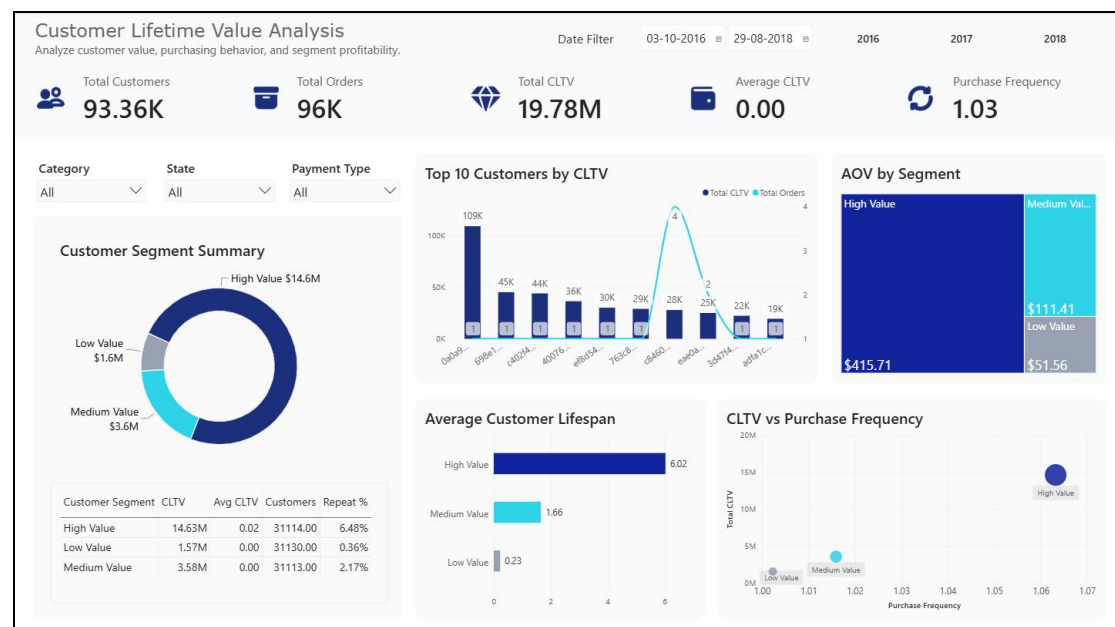
This dashboard focuses on overall customer value, purchasing behaviour and customer segmentation.

Main KPIs

- Total Customers: 93.36K
- Total Orders: 96K
- Total CLTV: \$19.78M
- Purchase Frequency: 1.03

Main visuals

- Customer Segment Summary
- Top 10 Customers by CLTV
- AOV by Customer Segment
- Average Customer Lifespan
- CLTV vs Purchase Frequency
- Segment-level analytical table



Key observations

- The High Value segment is the largest contributor to CLTV, generating approximately \$14.63M.
- High Value customers have a substantially higher AOV of \$415.71.
- High Value customers also show the longest average customer lifespan at approximately 6.02.
- The High Value segment has the highest repeat rate at 6.48%, compared with 2.17% for Medium Value and 0.36% for Low Value customers.
- The relationship between purchase frequency and CLTV highlights the importance of encouraging customers to purchase more frequently.
- The top-customer analysis identifies individual customers with significantly higher CLTV than the rest of the customer base.
- Total Customer are 93.36K with 96K orders depicting business's low repeat rate and high unique customers.

8. Key Business Insights

1. Revenue is strong, but repeat purchasing is a major opportunity

The business generates \$19.78M in revenue, but the overall repeat rate is only 3%. This suggests significant potential for increasing revenue through customer retention rather than relying only on new customer acquisition.

2. High Value customers drive the majority of customer lifetime value

The High Value segment contributes approximately \$14.63M CLTV, making it the most commercially important customer group.

3. Higher order value is strongly associated with higher customer value

High Value customers have an AOV of approximately \$415.71, compared with \$111.41 for Medium Value and \$51.56 for Low Value customers.

4. Customer lifespan is an important differentiation

High Value customers have an average lifespan of approximately 6.02, substantially higher than Medium Value customers at 1.66 and Low Value customers at 0.23.

5. Retention and repeat purchasing are critical growth levers

The difference in repeat rates between High Value and Low Value customers indicates that encouraging second and subsequent purchases could have a meaningful impact on CLTV.

6. Cohort retention declines as customer age increases

The cohort matrix demonstrates that customer retention generally falls across later cohort periods, highlighting the need for targeted post-purchase engagement and retention strategies.

9. Business Recommendations

Based on the analysis, the following actions can be considered:

1. Strengthen repeat-purchase campaigns

Develop targeted campaigns like personalized offers, reminder campaigns, post-purchase recommendations, limited time offers etc to move first-time customers toward their second purchase.

2. Protect High Value customers

High Value customers represent the largest CLTV contribution. To protect High Value customers programs like Loyalty programs, Personalized recommendations, VIP benefits, Early Access offers etc can be carried out.

3. Increase AOV among Medium Value customers

Medium Value customers represent an opportunity to move customers toward the High Value segment.

Potential strategies:

- Product Bundles
 - Cross-selling
 - Volume Discounts
 - Free-shipping Thresholds
 - Related product recommendations
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4. Reactivate Low Value customers

Low Value customers have the lowest AOV, lifespan and repeat rate.

Instead of treating all customers equally, targeted reactivation campaigns can be used to identify customers with potential for conversion into repeat buyers.

5. Monitor cohort retention continuously

Cohort analysis should be used as an ongoing KPI to determine whether new customer cohorts are becoming more valuable and more retained over time.

10. Conclusion

This project demonstrates how transactional customer data can be transformed into a practical customer analytics solution using Python and Power BI.

With Low retention rate and more unique buyers the analysis highlights a clear business opportunity: increase customer retention and purchasing frequency while protecting and expanding the High Value customer segment.

By combining cohort analysis with CLTV and customer segmentation, the business can move beyond simply measuring revenue and begin understanding which customers create long-term value and how that value can be increased.

Project Deliverables

This project includes:

- ✓ Project Requirements
 - ✓ Raw Olist E-commerce Brazilian Data
 - ✓ Cleaned Dataset Notebook
 - ✓ EDA Notebook
 - ✓ CLTV analysis Notebook
 - ✓ Cohort Retention Analysis Notebook
 - ✓ Power BI file
 - ✓ Dashboard Images
 - ✓ Python Charts
 - ✓ Executive Report
 - ✓ Business Insights and Recommendations
 - ✓ Project presentation
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